Q3) $x_1 + 2x_2 + 2x_3 + 4x_4 \leq 40$ $\quad \min z = 8x_1 + 6x_2 + 3x_3 - 2x_4$

$2x_1 - x_2 + x_3 + 2x_4 \leq 8$

$4x_1 - 2x_2 + x_3 - x_4 \leq 10$

$x_1, x_1, x_3, x_4 \geq 0$

Basic	z	x_1	x_2	x_3	x_4	s_1	s_2	s_3	Solution	
z	1	-8	-6	-3	2	0	0	0	0	
s_1	0	1	2	2	4	1	0	0	40	$\frac{40}{4} = 10$
s_2	0	2	-1	1	2	0	1	0	8	$\frac{8}{2} = 4$ ✓
s_3	0	4	-2	1	-1	0	0	1	10	$\frac{10}{-1} = -10$ ×

Basic	z	x_1	x_2	x_3	x_4	s_1	s_2	s_3	Solution
z	1	-10	-5	-4	0	0	-1	0	-8
s_1	0	-3	4	0	0	1	-2	0	24
x_4	0	1	$-1/2$	$1/2$	1	0	$1/2$	0	4
s_2	0	5	$-\frac{5}{2}$	1	0	0	1	1	14

$-2 - (-5 \times 1)$

$-2 + 1$ (+1)

-1 (-1)

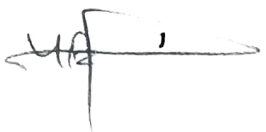
$-2 - \frac{1}{2}$

$\min z = 8x_1 + 6x_2 + 3x_3 - 2x_4 = -8$

~~$x_4 = 4$~~ $\quad x_1, x_2, x_3 = 0$

) no positive ✓

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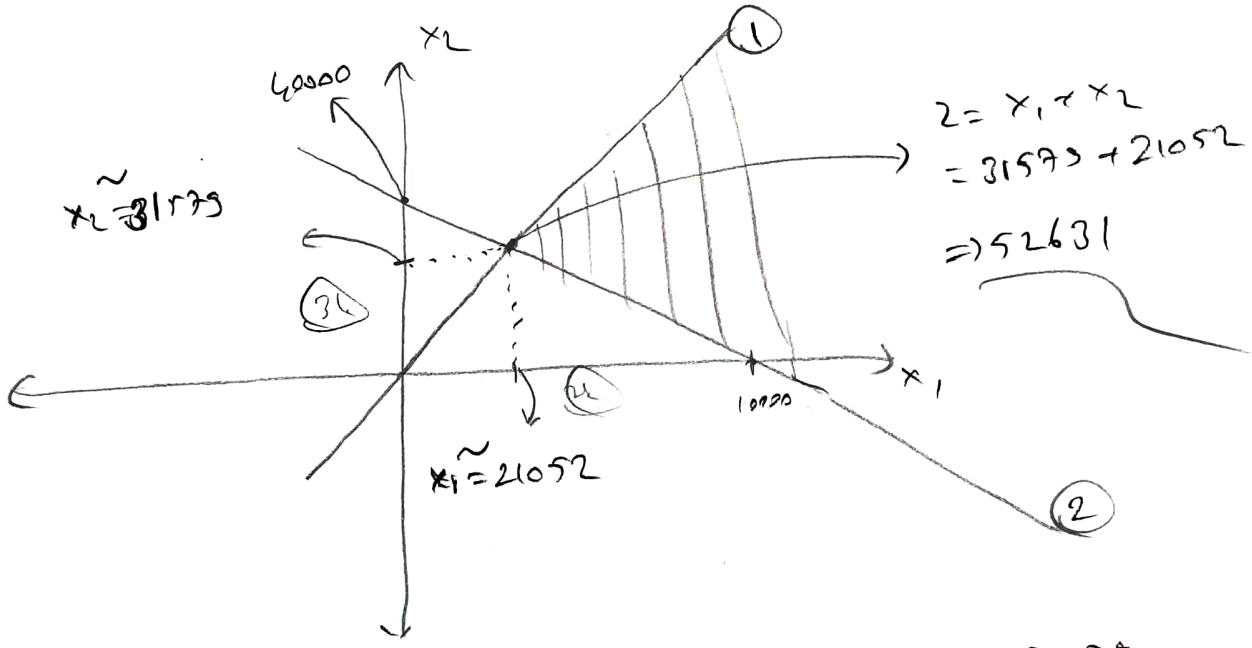
Q4) x_1: Amount invested in blue chip stocks
x_2: Amount invested in high tech stocks

minimize $Z = x_1 + x_2$

constraints:

$0.10 x_1 + 0.25 x_2 \geq 10000$ (1)

$x_2 \leq (x_1 + x_2) 0.6 \longrightarrow 0.4 x_2 - 0.6 x_1 \leq 0$ (2)



$Z = x_1 + x_2$
$= 31579 + 21052$
$\Rightarrow 52631$

$0.4 x_2 = 0.6 x_1$

$x_2 = \frac{3}{2} x_1$

$\frac{10 x_1}{100} + \frac{25}{100} \cdot \frac{3}{2} x_1 = 10000$

$20 x_1 + 75 x_1 = 2000000$

$95 x_1 = 2000000$

$400000 \mid 19$
$38 \quad\quad 21052...$
20
19
100
95
50

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